

Worldwide Microcontroller Revenue Share by Supplier

Company	2008		
	rank	\$ million	share
Renesas Technology	1	2770	20.1%
Freescale Semiconductor	2	1518	11.0%
NEC	3	1330	9.7%
Fujitsu	4	1065	7.7%
Infineon Technologies	5	983	7.2%
Microchip Technology	6	812	5.9%
STMicroelectronics	7	645	4.7%
Texas Instruments	8	601	4.4%
Atmel	9	511	3.7%
NXP Semiconductors	10	286	2.1%
Other		3229	23.5%
Total		13,749	—

Source: Databeans

control and safety features like airbags, ABS, electronic stability control (ESC) and advanced motor control. Luxury features like keyless entry, satellite radio and GPS-based navigation are also driving the use of microcontrollers in automobiles. There are 20-40 electronic control units (ECUs) in a car that uses MCU as the brain of the electronic circuit. Hybrid vehicles use emission control and fuel-efficiency control systems based on MCUs. "MCUs today have media-oriented systems transport (MOST), FlexRay, controller area network (CAN) and many more automotive buses embedded for easy automotive electronics design," informs Desmond Tan, senior marketing manager, South Asia/India, Fujitsu.

Medical electronics. The medical electronics industry is gaining momentum with increased longevity of the people. As health-care expenditure increases, there will be greater need for affordable medical care without compromising on quality. MCUs help bring down the size and cost of cumbersome medical equipment by enabling miniaturisation in medical devices. Portable healthcare devices using MCUs—like blood pressure monitors, glucometers and handheld ultrasound systems—have allowed healthcare at home, creating more value for consumers. Majority of the healthcare and fitness equipment

are microcontroller-based.

Industrial automation.

With shorter product life-cycles, there is increased need for flexibility and agility in the industrial environment. This has resulted in increased use of automation. Microcontrollers enable high-end automation by supporting complex feedback control systems, data acquisition systems and advanced sensors.

There are various industrial applications of MCUs, which include smart metering, HVAC and motor control. Smart metering systems help control energy usage in homes

as well as in industrial applications by reporting energy levels wirelessly and prompting the users about the costs being incurred and energy being spent. When equipped with a microcontroller, a smart meter can be configured to monitor other factors such as power strength and quality. MCUs help increase the power efficiency of the motors that go into the appliances. These can cut the power consumption by a decent amount or reduce the motor size significantly.

Consumer electronics and home appliances. Consumer electronics and home appliances have seen the most widespread use of microcontrollers. Everything that is attractive in consumer electronics—like touch screen, LED lighting or audio-video in your mobile phone—can be designed using microcontrollers. Home appliances and white goods—like washing machines, air-conditioners with energy saving mode, refrigerators with auto defrost and microwaves with various setting options—are made intelligent and feature-rich by the control logic driven by the microcontroller.

Technology benchmarks and trends

A microcontroller can actually be seen as integration of many technologies such as processor, memory and pe-

Vendor profiles

Atmel

Atmel offers both 8-bit and 32-bit AVR flash microcontrollers and ARM microcontrollers and application processors in the Indian market.

The Atmel AVR combines a code-efficient architecture for 'C' and Assembly programming with the ability to tune system parameters throughout the entire life-cycle of products. The 32-bit AVR UC3 product family is built on the high-performance 32-bit AVR architecture and optimised for highly integrated applications. The 32-bit AVR CPU includes cutting-edge features such as integer- and fixed-point DSP arithmetic, single-cycle multiply-and-accumulate instructions and single-cycle SRAM access.

The 8-bit/16-bit AVR XMEGA family delivers higher embedded performance combined with lower cost, smaller size and lower power. Apart from these, Atmel offers 8-bit megaAVR and 8-bit tinyAVR. The Atmel megaAVR family features self-programming flash for fast, secure, cost-effective field upgrades, while tinyAVR devices are optimised for a wide range of applications that require a small but powerful microcontroller. Atmel also offers MCUs based on ARM architecture.

Freescale

Freescale's 8-bit MCUs provides a wide range of low-power, highly-functional solutions for industrial, consumer and automotive markets, while 16-bit S12 and S12X MCUs provide high-performance control for automotive and industrial applications.

The S12X MCUs feature the innovative XGATE module, designed specifically to handle interrupt events without CPU intervention. As a result, the S12X controller has high-performance capabilities you would normally expect of a 32-bit controller. The 16-bit portfolio also includes a large family of digital signal controllers (DSCs), which combine microcontroller functionality with DSP performance. These are particularly suited for advanced motor control applications.

The 32-bit ColdFire embedded controller offers a range of performance and peripheral options, including one of the lowest-power 32-bit MCUs on the market. It is a cost-effective and easy-to-use 32-bit solution for your next consumer or industrial application.

ripherals. Any development in these technologies leads to an innovation in MCU. Some of the major technological changes seen in the MCU market are the move from 16-bit to 32-bit, introduction of ARM Cortex M series,